TRACE AND NORM OF AN ELEMENT READ OFF ITS MINIMAL POLYNOMIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The computational form of trace and norm used throughout the trace/norm section and its discriminant applications.

Editorial note. The label lemma-trace-and-norm-from-minimal-polynomial names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-characteristic-vs-minimal-polynomial (source lemma, worklist row 36): for a finite extension L over K and alpha in L with minimal polynomial P of degree d, the characteristic polynomial of the K-linear multiplication map m_alpha on L equals P^e, where e d = [L : K]
- Chapter definitions: Trace of an element is minus the coefficient of x with exponent n-1, and Norm is (-1)^n times the constant term, of the characteristic polynomial of the multiplication map, n = [L : K]

Source reference. Original reference 0BIG: sources/stacks-fields/pinned-original-source.tex, source lines 2616–2616. Prerequisite pointer invoked at 2656; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be a finite extension of fields. Let $\alpha \in L$ and let $P = x^d + a_1 x^{d-1} + \ldots + a_d$ be the minimal polynomial of α over K. Then*

$$\mathit{Norm}_{L/K}(\alpha) = (-1)^{[L:K]} a_d^e \quad \text{and} \quad \mathit{Trace}_{L/K}(\alpha) = -e a_1$$

where $ed = [L : K]$.

Proof. Follows immediately from Lemma 0BIG and the definitions. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.